Predicting Social Media Engagement

-The machine learning approach

Why do some posts go viral but others don’t?



LOW ENGAGEMENT



HIGH ENGAGEMENT

THE CHALLENGE

Problem: Decision Uncertainty

Choose the Right Platform

Businesses struggle to identify where their specific target audience is most active and receptive to content.

Choose the Right Content

Deciding between short-form video, images, or text posts is often based on intuition rather than data.

Understand the Audience

User preferences shift rapidly, making it difficult to maintain consistent engagement without predictive insights.

The Reality

Engagement is highly variable, making traditional marketing strategies risky and inefficient without data-driven models.

METHODOLOGY

Approach: Three Incremental Models

BASELINE

Model 0

Basic Features

- ✓ Platform (FB, IG, TikTok)
- ✓ Content Type (Video/Img)
- ✓ Basic Demographics
- ✓ Post Timestamp

COMPLEXITY: LOW

ENRICHED

Model 1

Content + History

- \+ Video Length / Text Count
- \+ Hashtag Analysis
- \+ Account History (Likes)
- \+ Posting Frequency

COMPLEXITY: MEDIUM

ADVANCED

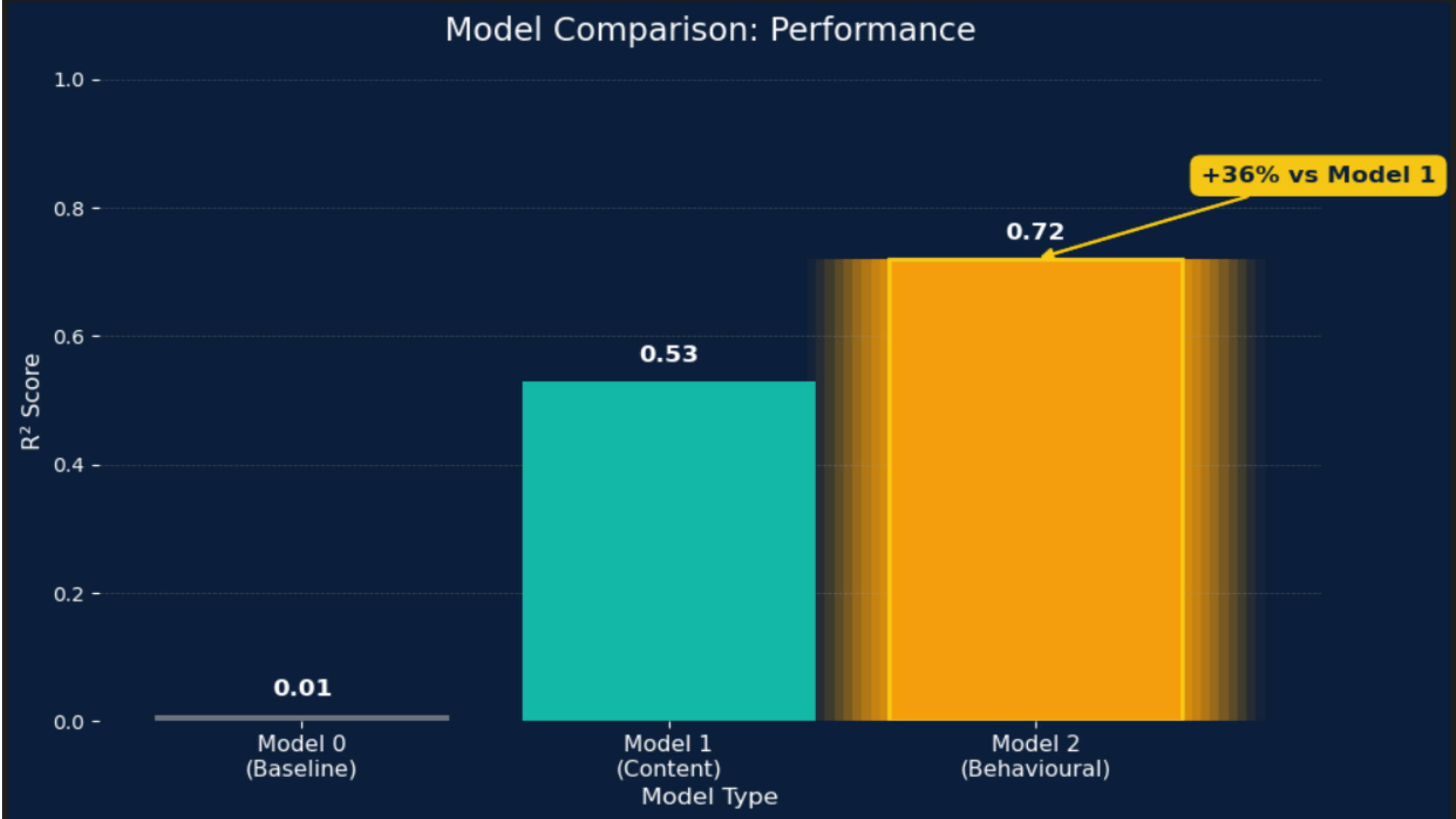
Model 2

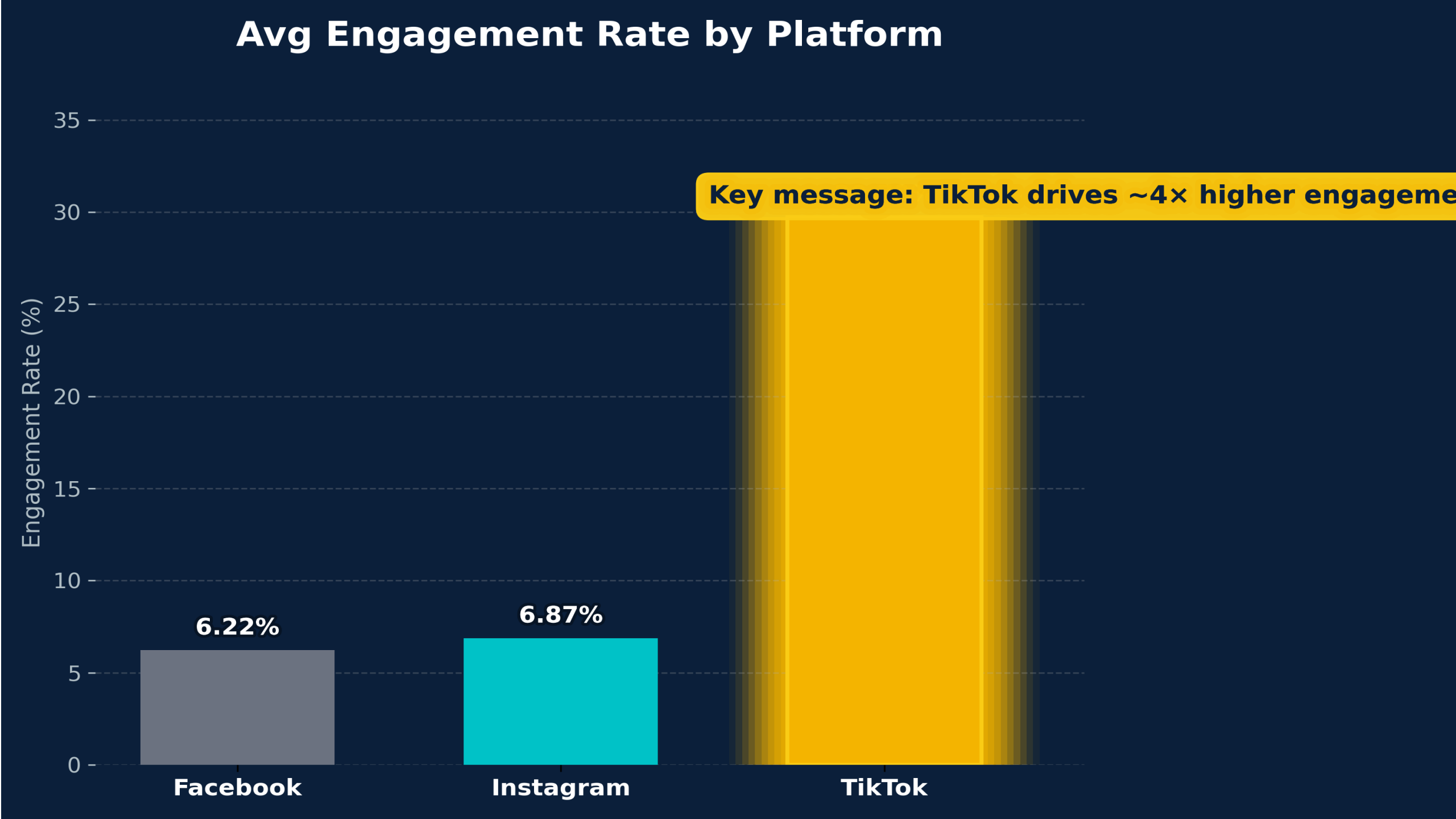
Behavioral + Deep

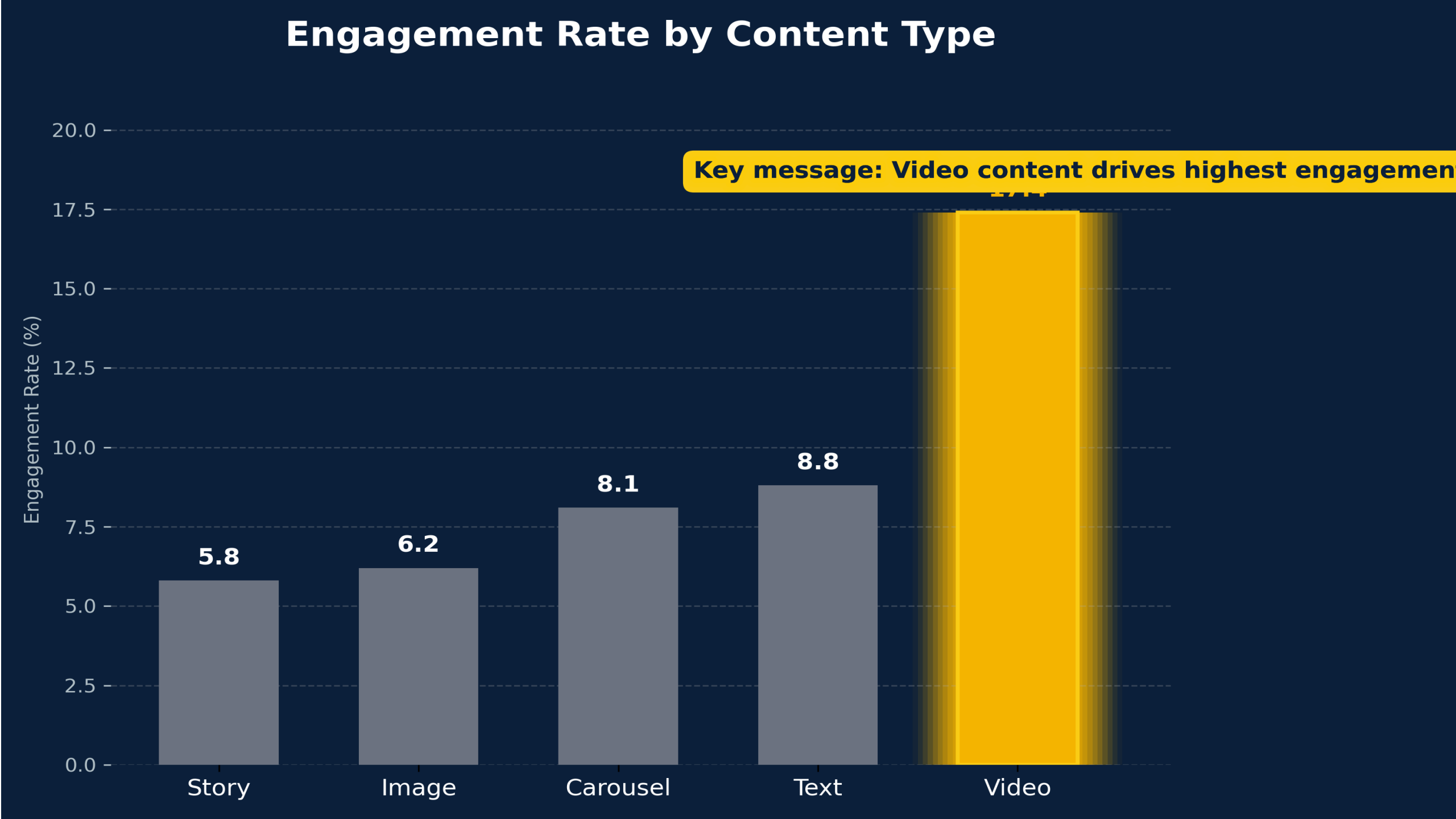
- ⚡ Session Interactions
- ⚡ Dwell Time & Re-visits
- ⚡ Follower Graph Data
- ⚡ Audience Segments

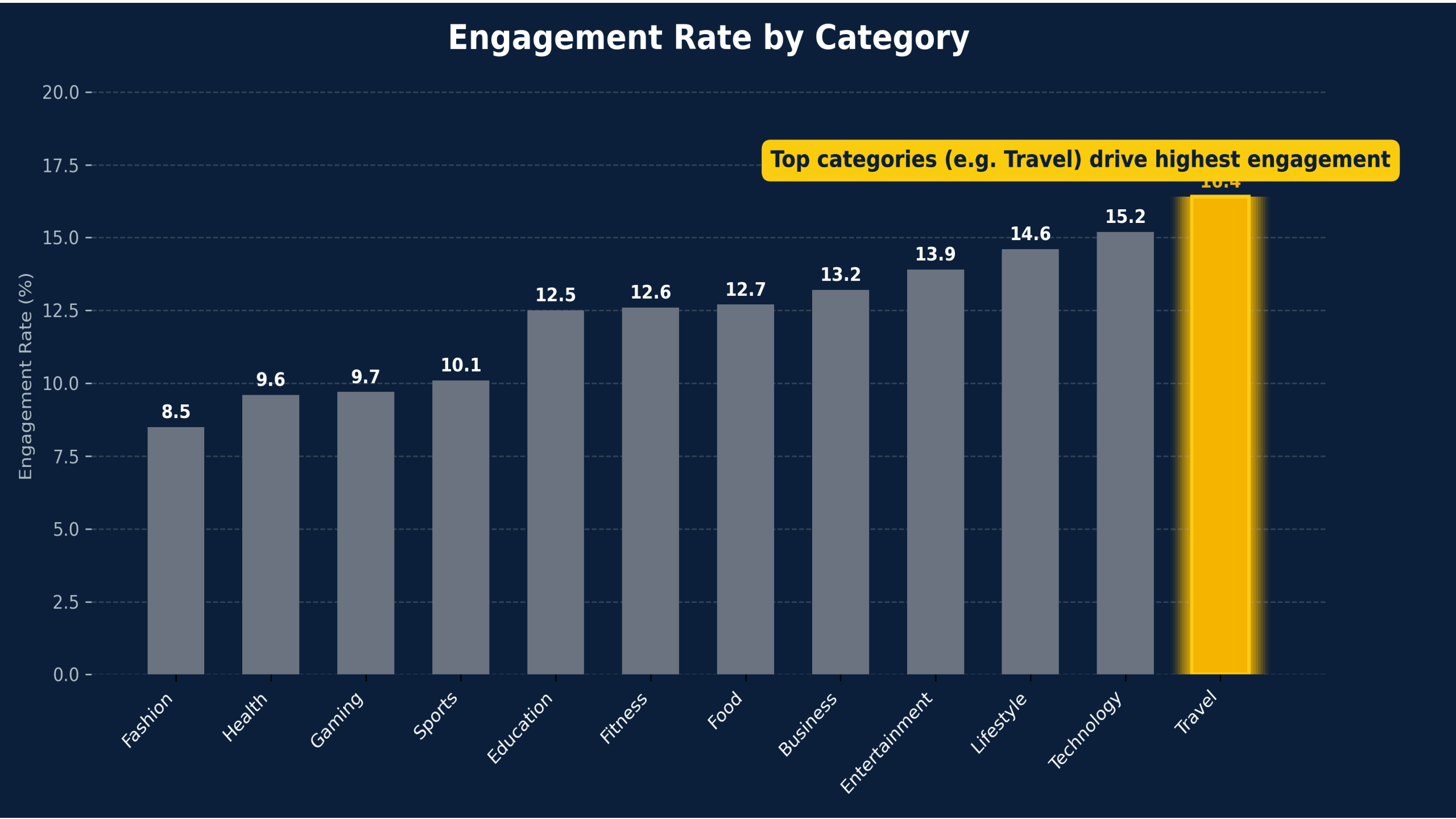
COMPLEXITY: HIGH

Our hypothesis: Adding behavioral features (Model 2) significantly reduces prediction error.


Model Comparison: Performance
1.0
0.8
0.6
0.4
0.2
0.0
R² Score
0.01
0.53
0.72
+36% vs Model 1
Model 0
(Baseline)
Model 1
(Content)
Model 2
(Behavioural)
Model Type


Avg Engagement Rate by Platform
Engagement Rate (%)
35
30
25
20
15
10
5
0
6.22%
6.87%
Facebook
Instagram
TikTok
Key message: TikTok drives ~4× higher engageme


Engagement Rate by Content Type
Key message: Video content drives highest engagemen
Engagement Rate (%)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
5.8
6.2
8.1
8.8
Story
Image
Carousel
Text
Video


Engagement Rate by Category
Top categories (e.g. Travel) drive highest engagement
Engagement Rate (%)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
8.5
9.6
9.7
10.1
12.5
12.6
12.7
13.2
13.9
14.6
15.2
Fashion
Health
Gaming
Sports
Education
Fitness
Food
Business
Entertainment
Lifestyle
Technology
Travel

DATA FEATURES

Feature Engineering & Selection

Content
BASE ATTRIBUTES

Static properties extracted directly from the post metadata and media file.

- Type
- Length
- # Hashtags
- Post Time
- Audio ID

HIGHEST IMPACT

Behavioral
DYNAMIC SIGNALS

User interaction patterns captured during the first hour of exposure. These features proved most predictive.

- Interactions/Session
- Dwell Time
- Re-visit Rate
- Share Velocity
- Completion Rate

Demographic
AUDIENCE PROFILE

Characteristics of the account's existing follower base and target region.

- Gender
- Age Group
- Region
- Follower Count
- Language

MODEL 1 FEATURES MODEL 2 FEATURES (ADDED)

RESULTS ANALYSIS

Key Findings

High Predictability

Engagement is predictable with the right feature set. Our best model (Model 2) achieved a strong R^2 of 0.72.

TikTok Dominance

TikTok consistently outperforms other platforms in engagement metrics across all data segments in our study.

Behavioral Drivers

Behavioral data (interactions, dwell time) explains the most variance, proving more predictive than basic demographics.

Strategic Implication

While content quality matters, how users interact with it (behavior) is the strongest indicator of future virality.

SUMMARY

Project Conclusion

Model 0: Failed

Simple features (Age, Gender, Platform) were insufficient for prediction. The model failed to capture the complexity of user interest, resulting in near-random guessing.

Predictive Power $R^2 \approx 0.01$

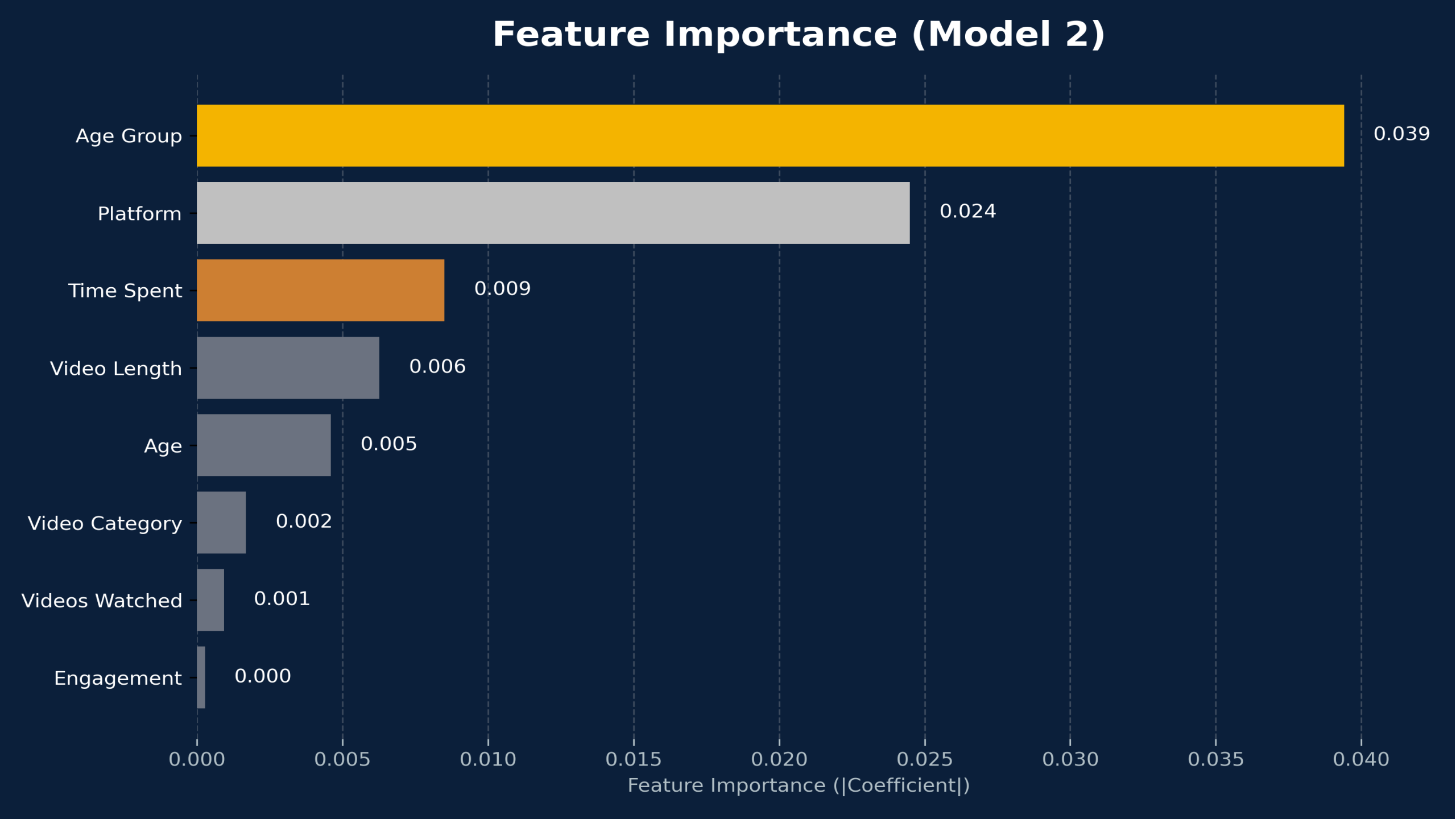
Model 2: Succeeded

Integrating **Behavioral** (interactions) and **Demographic** features significantly improved accuracy. Understanding how users interact is key.

Predictive Power $R^2 \approx 0.72$

Engagement is predictable.

When modeled with behavioral data, uncertainty is drastically reduced.


Feature Importance (Model 2)
Age Group
Platform
Time Spent
Video Length
Age
Video Category
Videos Watched
Engagement
0.039
0.024
0.009
0.006
0.005
0.002
0.001
0.000
0.000
0.005
0.010
0.015
0.020
0.025
0.030
0.035
0.040
Feature Importance (|Coefficient|)